2024 Spring 生化学 2 English, Homework#2 (250324)　　dead line: 23:59, March 31, 2025

学号:

姓名:

Q1. 判断题（请在括号里画√或 X, 1 分/题, 共 40 分）

(1) Almost all naturally occurring unsaturated fatty acids, the double bonds are in the cis configuration. (　　)
(2) If the carbon number is same, melting temperature for saturated fatty acids is higher than that of unsaturated one (　　)
(3) All the three fatty acids in triacylglycerols are same (　　)
(4) Phospholipids contain PO_4 in their polar head (　　)
(5) All phosphorlipids's head-group alcohol is same (　　)
(6) Lipids in achaea is same to that of E. coli (　　)
(7) Sphingomyelin contains glycerol backbone (　　)
(8) Sphingolipids contain ceramide (　　)
(9) Ganglioside contains oligosaccharide (　　)
(10) Phospholipase C split the C-1 of glycerol (　　)
(11) Sterols contain carbon rings (　　)
(12) PIP2 is hydrolyzed to diacylglycero and IP2 (　　)
(13) Eicosanoid derivatives go around the whole body (　　)
(14) Some Vitamins are derivative of isoprene (　　)
(15) Vitamin A binds to opsin after chemical modification (　　)
(16) Light converts sterol to Vitamin D (　　)
(17) Human can make Vitamin K by themselves (　　)
(18) Coenzyme Q works as electron carrier (　　)
(19) Lipid can make color in plant (　　)
(20) Lipids are normally extracted from the sample with inorganic solvent (　　)
(21) K+ ion itself is easy to cross the cell membrane (　　)
(22) In animal, cholesterol is the major components of plasma membrane (　　)
(23) Cardiolipin is a major components of Golgi membrane (　　)
(24) In the Micelle, cross section of head equals that of side chain (　　)
(25) SDS (Sodium dodecyl sulfate) forms open bilayer (　　)
(26) Nanodisc is made of lipids and nucleic acids (　　)

(27) Lipid contents of inner- and outer leaflet is same ( )

(28) Phosphatidylserin is mostly in outer leaflet ( )

(29) Integral protein can be extracted from membrane using detergents ( )

(30) Some membrane proteins associate with membrane through covalently attached lipid anchors ( )

(31) Hydropathy plots is the index to identify hydrophobic and hydrophilic part of protein ( )

(32) If protein cross the membrane with a-Helix, only 7-9 amino acids are enough from one leaflet to the other leaflet ( )

(33) GPI anchor protein is in inner leaflet and can be extracted from membrane using phospholipase C ( )

(34) In the case of E. coli, even if the culture temperature change, the lipid composition does not change. ( )

(35) Uncatalyzed transbilayer ("flip-flop") diffusion is quite fast ( )

(36) Flippase moves PE from inner leaflets to outer leaflets ( )

(37) Scramblase use ATP to work ( )

(38) Glycerophospho lipids are enriched in Rafts ( )

(39) t-SNARE is on the vesicle ( )

(40) Integrin is homodimer ( )

Q2. Please draw the structure of the following lipid 18:2($\Delta^{9,12}$) (2 分)

Q3. Please explain the character of Fluid Mosaic Model (4 分), and describe two experiments precisely that support the Model (各 2 分、共 8 分)